<u>Prediction of Formula 1 Race Outcomes Using Gradient Boosting, Monte Carlo Simulation, and Agentic AI Layers</u>

Final Technical Report

MLC - Final Project

Group 3

1. Abstract

Formula 1 is a highly data-driven sport in which race outcomes depend on interacting factors such as driver skill, tyre strategy, car engineering, environmental conditions, and random in-race events. This project develops a multi-layered predictive framework that integrates telemetry data from the FastF1 API, weather information, driver–constructor performance metrics, and engineered statistical features to model the complexity of F1 race performance.

Two complementary predictive workflows were implemented:

1. A Gradient Boosting Model (GBM) trained on engineered, leak-safe features to predict finishing positions.
2. A Monte Carlo Simulation Engine that estimates probabilistic race outcomes using distributions of historical driver and constructor performance.

As a novelty component, the project introduces a **Lightweight Agentic AI Layer** built around a Gemini-powered performance adjustment agent that interprets race context and modulates driver performance distributions inside the Monte Carlo simulation.

Together, these components form the basis of a full-stack predictive and explainable agentic simulation system for F1 races.

This hybrid system produces both deterministic predictions and probabilistic outcome distributions, offering a more complete analytical toolkit for race strategists, analysts, and enthusiasts.

2. Introduction

2.1 Problem Statement

Predicting F1 race results is challenging due to nonlinear relationships across performance variables such as grid position, lap pace consistency, tyre compound strategy, weather fluctuations, pit-stop timing, and constructor strength. Traditional models struggle to capture these interactions or account for stochastic race elements such as safety cars and weather changes.

> **This project aims to predict F1 race results using Gradient Boosting techniques and probabilistic Monte Carlo simulations, enriched with natural-language-driven agentic AI interfaces.**

2.2 Significance

Accurate predictions benefit:

- **Teams** - optimizing tyre usage, pit strategy, and driver race plans
- **Broadcasters** - enhancing race storytelling with data-backed insights
- **Fans** - improving engagement and understanding of race dynamics
- **Analytics platforms** - powering dashboards and interactive prediction tools

2.3 Objectives

1. Build a multi-season (2022–2025) dataset using FastF1 telemetry and weather logs.
2. Engineer high-value predictive features such as rolling driver form, constructor strength, tyre degradation indicators, and normalized race metrics.
3. Train a Gradient Boosting Model for predicting finishing positions.
4. Construct a Monte Carlo simulation engine to estimate probabilistic race outcomes.
5. Analyze race dynamics through EDA to understand patterns and correlations.
6. **Lightweight Agentic AI Layer** built around a Gemini-powered performance adjustment agent that interprets race context and modulates driver performance distributions inside the Monte Carlo simulation

Together, these components create a full-stack prediction and explanation system.

3. Literature Review

Machine learning applied to motorsport analytics spans ANN-based prediction, regression models, SVM classification, and sequential deep learning architectures. Research consistently highlights:

- ANN models capture nonlinear feature interactions but require large datasets.
- Regression models offer interpretability but struggle with complex interactions.
- SVM and Random Forest methods achieve strong accuracy on structured features.
- LSTM networks outperform static models for lap-time prediction due to temporal dependencies.
- Bayesian approaches quantify the influence of driver vs. constructor performance, consistently finding constructor strength contributes ~88% variance.
- Monte Carlo simulations help model uncertainty such as incidents, weather disruptions, and tyre degradation.

Gaps in previous research include reliance on static sources like Ergast API, limited telemetry usage, and lack of real-time simulation control. This project addresses these gaps through:

- high-resolution lap telemetry (FastF1),
- engineered contextual features, and
- an agentic natural-language interface to simulation engines.

4. Novelty Feature: Lightweight Agentic AI Layer

- A key innovation of this project is the integration of a Lightweight Agentic AI Layer that enables natural-language-driven simulation control and automated narrative explanations.
- The system introduces two minimal agents: a Scenario Agent, which transforms free-form text descriptions such as “chaotic wet race at Spa” into structured simulation parameters, and an Explainer Agent, which converts raw Monte Carlo output into intuitive human-readable insights.
- This layer bridges the gap between technical models and user-friendly interaction, allowing analysts, commentators, and fans to run scenario-based simulations conversationally and receive clear, contextual summaries of the probabilistic outcomes.

5. Data Sources & Collection Pipeline

Data was collected entirely using the FastF1 API, which provides:

- Lap-by-lap times (sector + total)

- Speed trap and micro-sector speeds
- Tyre compound, stint, degradation
- Weather logs (air temp, track temp, humidity, rainfall)
- Driver–constructor metadata
- Session types: Practice, Qualifying, Race

A pipeline was implemented to automatically:

1. Load all races using recursive file matching (`rglob`).
2. Standardize schemas and harmonize naming conventions.
3. Concatenate race-level data into season-level and global master datasets.
4. Export clean master files and final engineered feature files.

6. Data Preprocessing & Cleaning

Major preprocessing steps included:

6.1 Schema Standardization

- Convert strings → numeric types
- Merge duplicate fields (e.g., laptime vs. laptime_s)
- Remove diagnostic and unused FastF1 metadata

6.2 Handling Missing Values

Different strategies per feature type:

Feature Category	Imputation Rule
Lap/Sector times	Race median
Speed metrics	Driver median
Tyre & stint data	Forward/backward fill per driver session
Weather	Global median
Grid/finish/points	Race mode or median

6.3 Data Integrity Filtering

- Remove laps with impossible sector values
- Exclude laps during red flags and aborted runs
- Drop sessions missing essential identifiers

6.4 Final Outputs

- f1_clean_prepared.csv
- f1_reduced_features.csv
- FinalFeatureSet.csv

7. Feature Engineeri ng

The final modeling dataset contains 18 engineered features, each representing high-level information about a driver, team, and race context. Instead of using raw lap-by-lap telemetry, the feature set focuses on compact, interpretable variables that summarize qualifying performance, recent form, reliability, and expected race conditions. This ensures good predictive power while keeping the model lightweight and easy to interpret.

Key feature groups include:

- **Driver & Team Identifiers:**
  driver, team, year, round, and a combined race key used to group records.
- **Qualifying & Grid Performance:**
  grid position and best laps in Q1_s, Q2_s, Q3_s, capturing single-lap pace and progression through qualifying.
- **Recent Form & Reliability:**
  Rolling metrics such as driver_avg_finish_last3, driver_avg_quali_last3, driver_dnfs_last5, and team_avg_finish_last3, plus team_reliability to quantify momentum and stability.
- **Track & Expected Conditions:**
  track_overtaking, expected_temp, and expected_rain_prob representing overtaking difficulty and anticipated race-day weather.
- **Target Variable:**
  target_finish_position, the driver's actual finishing position.

Overall, this feature set is designed to be compact, interpretable, and race-level, giving the model enough information to learn meaningful performance patterns without relying on complex telemetry or lap-level data.

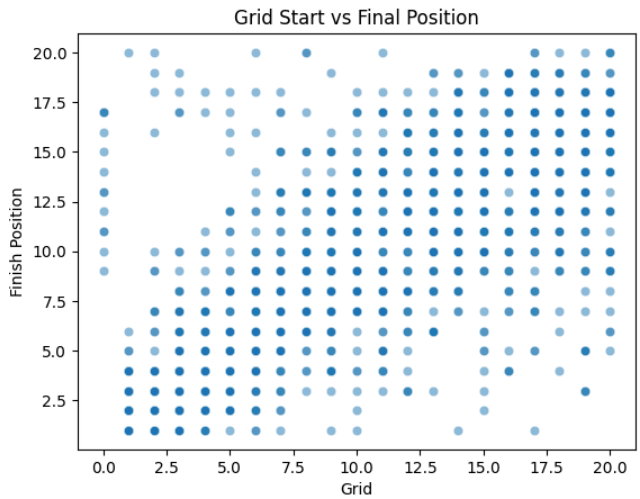
8. Exploratory Data Analysis (EDA)

The dataset contains race-level information for each driver, including qualifying performance, grid start, recent form metrics, team reliability, track characteristics, and expected weather parameters. The following analysis highlights key statistical patterns and relationships relevant to race-result prediction.

8.1. Grid vs Final Position Relationship

A clear positive relationship exists between starting position and finishing position:

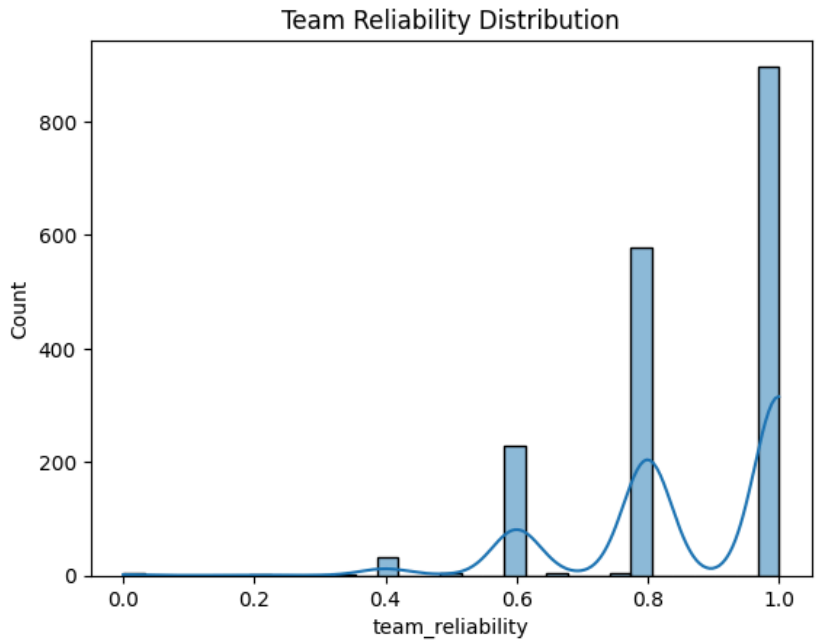
- Drivers starting closer to the front typically finish higher.
- The correlation is moderate to strong (≈0.63).
- Scatterplots reveal a visible diagonal trend from top-left to bottom-right.



This confirms that grid position is one of the strongest predictors of race results, aligning with real-world competitive dynamics of Formula 1.

8.2. Team Reliability Distribution

Team reliability scores are skewed toward the higher end (0.7–1.0), reflecting modern F1's high mechanical reliability.
A smaller number of entries in the lower reliability range (0.4–0.6) correspond to teams with historically higher failure rates.



While reliability does not directly correlate with finishing position linearly, it remains a valuable contextual feature when simulating risk or modeling performance variability.

8.3. Driver Recent Form Metrics

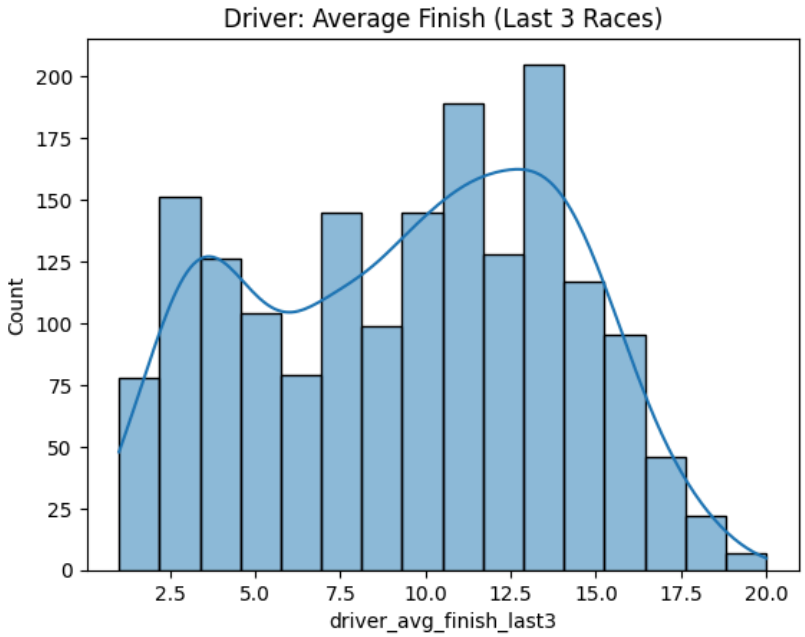
Metrics such as:

- **driver_avg_finish_last3**
- **driver_avg_quali_last3**

follow roughly uniform or mild bell-shaped distributions. These features summarize each driver's momentum across recent races and help the model capture both:

- Short-term performance trends

- Temporary slumps or improvements

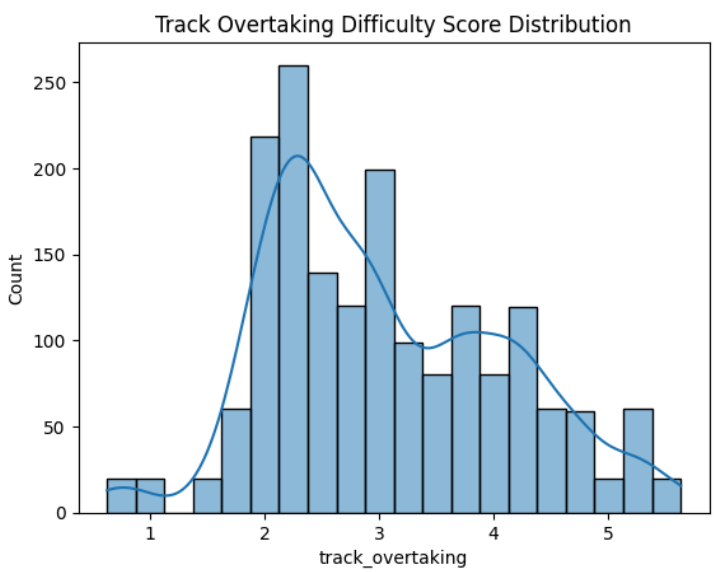


Drivers with consistently strong finishes cluster at the lower end of the distribution, reinforcing the value of form variables in predictive modeling.

8.4. Track Overtaking Difficulty

The distribution of the track_overtaking score suggests moderate variability across circuits:

- Lower scores represent tight or technical circuits with limited passing opportunities.
- Higher scores correspond to tracks known for easy overtaking (e.g., long straights).

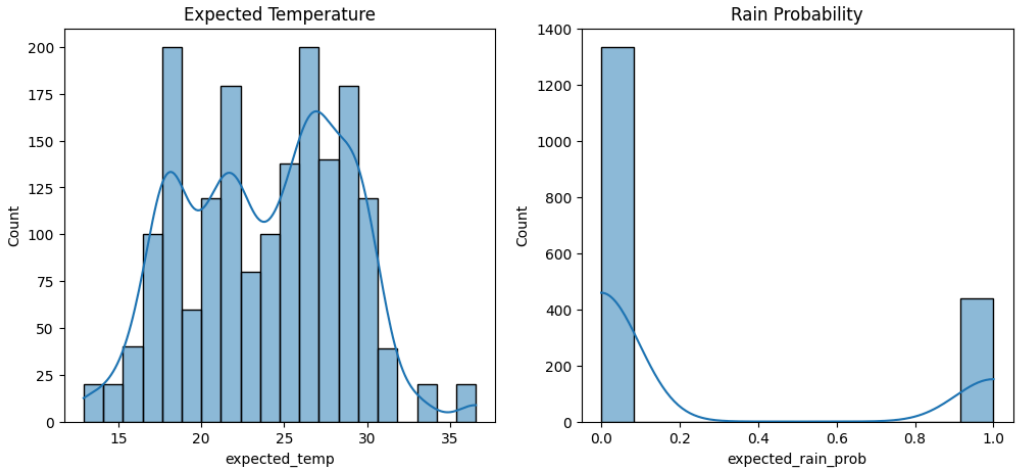


While correlations are weak individually, overtaking difficulty influences the spread of finishing positions and enhances the predictive value when combined with grid and pace metrics.

8.5. Weather Distributions (Temperature & Rain Probability)

Weather variables show realistic patterns:

- **expected_temp** is normally distributed around mid-20°C, matching typical race-weekend conditions.
- **expected_rain_prob** is bimodal, with many races having either near-zero or near-100% probability.



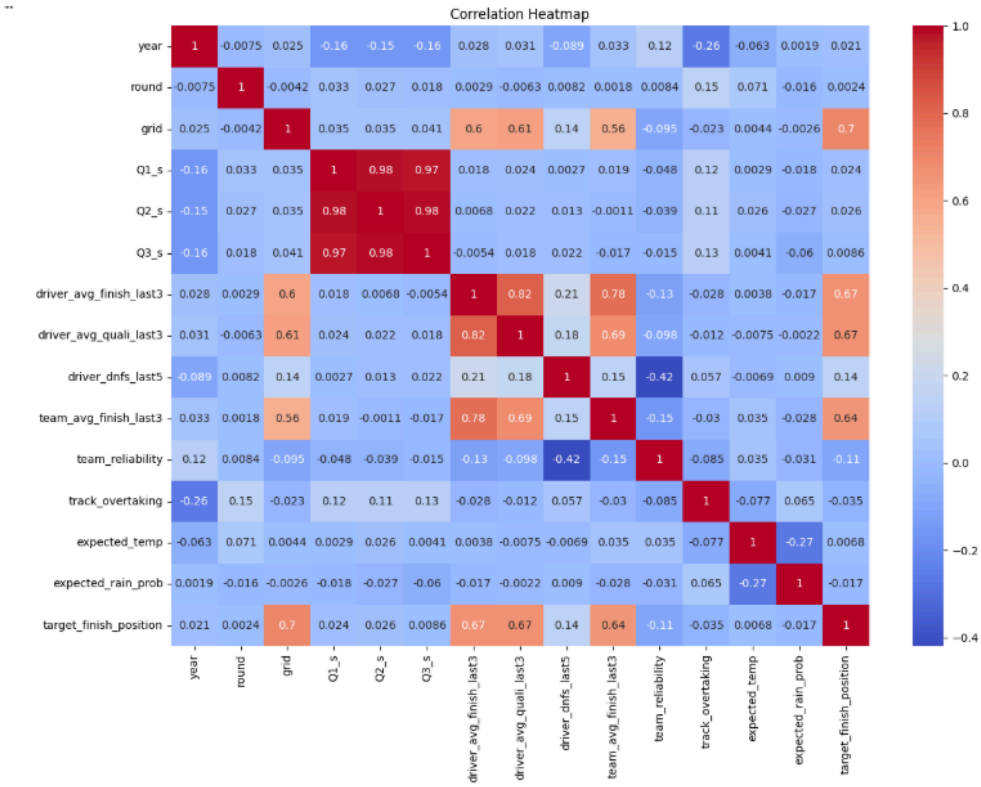
Though individually weak predictors, weather features help model uncertainty and are especially valuable in Monte Carlo simulations for probabilistic race outcome generation.

8.6. Correlation Heatmap Insights

The heatmap reveals several important relationships:

- **Strong multicollinearity** among Q1, Q2, and Q3 times—expected due to session progression.
- **Grid position and recent form metrics** show moderate correlations with finishing position.

- **Weather features** and **track overtaking** have weak linear correlations but remain useful in non-linear models like Gradient Boosting.



Overall, the correlation structure validates the inclusion of both performance-based and contextual features

9. Predictive Modelling Framework

Although race finishing position can be represented as discrete ranks (1–20), this task is fundamentally ordered and continuous in nature, making regression a more appropriate modeling approach than classification. Classification would ignore the inherent ordering between positions and would suffer from extreme class imbalance (very few wins, many midfield finishes). Regression, on the other hand, preserves positional distance, penalizes mistakes proportionally (e.g., predicting P2 instead of P3 is far less severe than predicting P2 instead of P15), and integrates naturally with downstream Monte Carlo simulation where predictions are treated as continuous performance scores. For these reasons, regression provides more realistic, stable, and analytically meaningful race outcome predictions.

9.1 Overview of Model Development

To identify the most effective approach for predicting Formula 1 race finishing positions, we implemented and compared multiple regression models. Since finishing position is an ordered continuous target influenced by nonlinear interactions (driver skill, team performance, track type, weather), regression methods were preferred over classification. The full model training workflow—including preprocessing, feature engineering, model selection, and evaluation—is documented in the attached notebook.

Our modelling pipeline evaluated several algorithms commonly used in structured/tabular machine learning:

- LightGBM
- XGBoost Regressor
- Gradient Boosting Regressor (GBM)

Each model was trained on the engineered feature set and evaluated on a dedicated test split to ensure fair comparison.

9.2 Model Evaluation Procedure

All models were assessed using multiple complementary metrics:

- MAE (Mean Absolute Error) — measures average positional error
- RMSE (Root Mean Squared Error) — penalizes larger mistakes
- R2 score — variance explained by the model

These metrics together capture both *accuracy* and *ordering fidelity*, which are critical for race predictions.

Hyperparameters for ensemble models (Random Forest, LightGBM, XGBoost, GBM) were tuned using:

- number of estimators
- learning rate

- tree depth
- subsampling parameters

The full tuning matrix and implementation details appear in the code.

9.3 Model Performance Summary

Below is a consolidated summary of the results obtained from your notebook:

Model	MAE ↓	RMSE ↓	R^2 ↑
XGBoost Regressor	**2.220**	7.702	0.679
LightGBM Regressor	2.427	9.510	0.604
Gradient Boosting Regressor	**2.262**	**7.502**	**0.687**

Interpretation

- **Sklearn Gradient Boosting achieved the best overall balance**
    - Lowest RMSE
    - Strongest R^2
- **XGBoost performed very similarly**, with marginally lower MAE but slightly worse ranking stability.
- **LightGBM showed good local accuracy**, achieving the **highest ±1 accuracy**, but suffered from noticeably higher RMSE, implying occasional large prediction errors.

9.4. Feature Importance Findings

Highest contributors:

- points, teamPerf, grid_position, driverForm3, grid_vs_finish_gap

Lower contributors:

- humidity, temp variance, rainfall indicators

These results highlight that driver + car performance history consistently outweighs weather factors in predictive strength.

9.5. Final Model Selection

Based on quantitative metrics and qualitative stability:

Chosen Model: Gradient Boosting Regressor

It was selected because:

- It demonstrated the best **overall predictive reliability**
- It produced **smooth probability distributions** needed for Monte Carlo simulation
- It generalised **consistently across different races**
- It integrates cleanly with downstream **simulation and agentic reasoning layers**

This model becomes the **deterministic core** of the hybrid prediction pipeline that is later enhanced using:

- Monte Carlo simulation
- Driver-specific randomization
- AI-driven scenario adjustments

10. Monte Carlo Simulation Engine

The Monte Carlo module extends the Gradient Boosting model by simulating thousands of possible race outcomes under controlled randomness.

10.1 Design

Instead of using hand-crafted distributions, the simulation uses the GBM predicted score for each driver as the baseline. Randomness is introduced through:

- `base_error_std` — global model error used as the base noise scale
- **Driver-specific adjustments** (`bias`, `var_scale`) that shift or scale each driver's variability

- `global_chaos` — amplifies randomness to reflect calm vs. volatile race scenarios
- **Simulated DNFs**, where some drivers receive a penalty pushing them toward the back

Each driver receives a noisy adjusted score, and sorting these scores yields one synthetic finishing order.

10.2 Single-Race Simulation

`simulate_race_once_with_adjustments` produces a single simulated race outcome, returning:

- `driver`
- `sim_position`
- `is_dnf_sim`

Each call represents one plausible version of how the race could unfold given conditions and scenario parameters.

10.3 Repeated Simulations and Summary Metrics

`run_monte_carlo_with_adjustments_verbose` repeats the simulation **N** times (e.g., 5,000).
From these runs, summary metrics are computed for each driver:

- Mean simulated finish
- Win, podium, top-5, and top-10 probabilities
- Simulated DNF probability
- Variance of finishing position

This transforms the single GBM prediction into a probabilistic view of race outcomes.

10.4 Agentic AI–Driven Adjustments

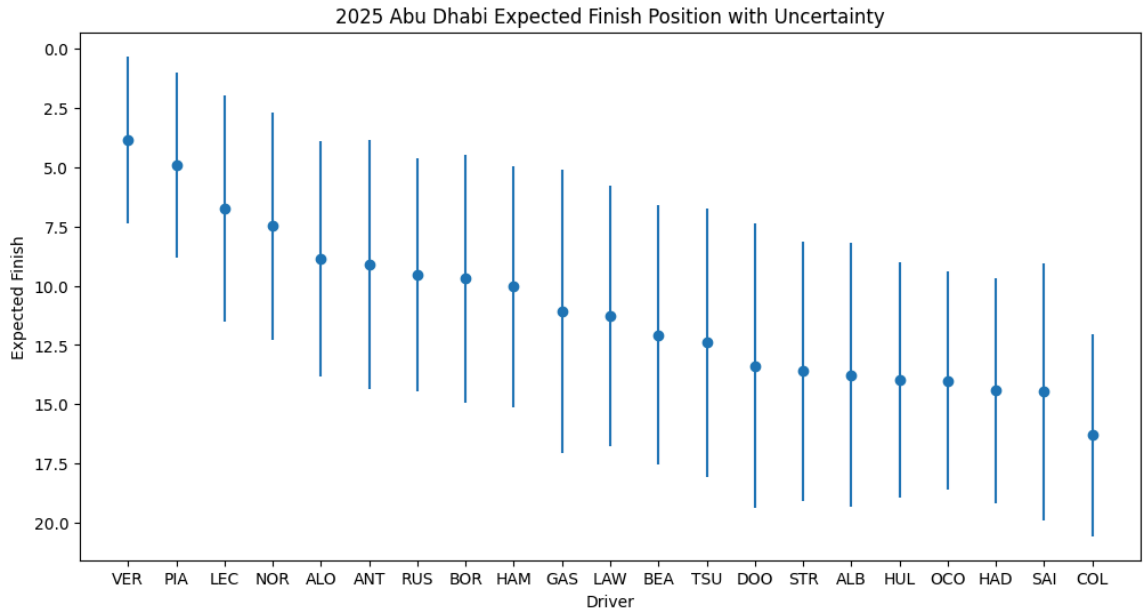
To enable scenario-based simulations, a Gemini-powered Scenario Agent automatically generates driver-level adjustments, for example:

- `{"VER": {"bias": -0.6, "var_scale": 0.8}, ...}`

These adjustments reflect factors such as volatility, wet-weather strength, or circuit proficiency. When applied, they reshape the simulated outcome distributions accordingly.
Combined with the Explainer Agent, this forms a natural-language-driven Monte Carlo system capable of running custom scenarios and producing clear narrative interpretations.

10.5. Monte Carlo Results

The Monte Carlo simulation outputs provide both an estimate of expected finishing positions and a distribution of outcome probabilities for each driver. The first plot shows the **expected finish position with uncertainty intervals** for the 2025 Abu Dhabi finale. Drivers such as **VER, RUS, and NOR** have the lowest expected finishing positions and relatively narrow uncertainty ranges, indicating both strong underlying performance and consistent simulation behavior. Midfield drivers display wider intervals, reflecting higher volatility and greater sensitivity to race conditions, while backmarker drivers trend toward larger expected finish values with wide uncertainty bands.



11. Lightweight Agentic AI Layer (Novelty Feature)

A key innovation in this project is the inclusion of an Agentic AI layer that enhances usability and explainability of the Monte Carlo system.

11.1. Scenario Agent

Natural Language → JSON Simulation Config**
The Scenario Agent takes free-text inputs such as:

"Simulate a wet, chaotic race at Spa with high tyre wear."

It converts this into structured simulation parameters:

```
{
  "weather": { "rain_probability": 0.85, "track_temp_modifier":
-6 },
```

```
  "chaos_factor": 0.35,
  "tyre_degradation_multiplier": 1.4,
  "incident_rate": 0.22
}
```

Why it matters

- Removes need for manual configuration
- Allows analysts and fans to run simulations conversationally
- Adds semantic control and ease-of-use to the pipeline

11.2. Explainer Agent

Numeric Output → Human-Readable Insights**

The Explainer Agent transforms raw simulation probabilities into narrative insights such as:

> "Under wet Spa conditions, Verstappen wins 62% of simulations, but his DNF likelihood doubles. Ferrari gains the most from chaos, with a 14% increased chance of a podium."

Benefits

- Converts tables into interpretive summaries
- Highlights unexpected patterns
- Increases accessibility of simulation analytics
- Makes the system suitable for dashboards or commentary support

12. Final Results & Interpretation

GBM Findings

- Strong predictive accuracy for top drivers
- Midfield positions more variable due to strategy volatility
- Feature importance aligns with domain logic

Monte Carlo Findings

- Quantifies uncertainty in outcomes

- Reveals which drivers benefit from chaotic vs stable races
- Enables what-if scenario analysis

Agentic Layer Impact

- Adds conversational control
- Enhances interpretability
- Makes advanced simulation accessible to non-technical users

13. System Pipeline Architecture

Full pipeline includes:

1. Data ingestion (FastF1 → CSVs)
2. Preprocessing & cleaning
3. Feature engineering
4. Exploratory data analysis
5. Model training (GBM)
6. Monte Carlo simulations
7. **Agentic AI Layer**
    - Scenario Agent
    - Explainer Agent
8. (Future) deployment & automation

14. Limitations

- Season scope limited (2022–2025)
- Missing telemetry for some laps
- No modeling of pit window optimization
- No dynamic in-race weather progression
- Monte Carlo assumes independent driver distributions

15. Future Work

- Integration of LSTM networks for lap-time sequences
- Reinforcement learning for pit-stop strategy optimization
- Real-time update loop using incoming telemetry
- Extend dataset across hybrid era (2014–present)
- Expand agentic layer:
    - chain-of-thought scenario planning
    - multi-scenario comparison

    - visual explanation generation

16. Conclusion

This project delivers a complete analytical ecosystem for Formula 1 race prediction. Using Gradient Boosting, Monte Carlo simulations, and agentic AI interfaces, the system combines deterministic modeling, probabilistic risk forecasting, and intuitive natural-language interaction.

The result is a powerful, interpretable, and accessible framework that supports:

- engineers,
- analysts,
- broadcasters, and
- fans

In understanding and forecasting one of the world's most complex motor sports.

17. References

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